

Brownian Bridge Diffusion Models (BBDM): Complete Technical Survey

Based on Source Materials

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Abstract

This report provides a complete technical survey of Brownian Bridge Diffusion Models (BBDM) based entirely on the source materials. All content, definitions, formulas, derivations, examples, and comparisons are taken directly from the provided file without addition. The survey covers: (1) the problem BBDM is solving - image-to-image translation between two domains; (2) the traditional approach of conditional generation and its limitations; (3) the central idea of constructing a stochastic bridge between source and target domains; (4) the Brownian Bridge mathematical formulation including mean, variance, and endpoints; (5) the difference between BBDM and linear interpolation; (6) latent space operation with VQGAN; (7) DDPM vs BBDM comparison; (8) the variance schedule and parameter s ; (9) Equation (3), (4), (5), (6), (8), (9), and (14) derivations; (10) the forward bridge equation $x_t = (1 - m_t)x_0 + m_ty + \sqrt{\delta_t}\epsilon$; (11) the reverse process and sampling algorithm; (12) why the network predicts displacement rather than pure noise; (13) training vs inference distinction; (14) accelerated sampling with $S = 200$; (15) the conceptual difference between conditional diffusion and BBDM; (16) the complete training and sampling algorithms; and (17) the key intuition that source and target are endpoints of the stochastic bridge rather than source being merely a condition.

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1 Introduction: What Problem is BBDM Solving?

Suppose we have two image domains:

- Domain A = sketches
- Domain B = photographs

We want:

$$x \in A \longrightarrow y \in B \quad (1)$$

For example:



More generally:

$$x \sim q_A(x), \quad y \sim q_B(y) \quad (2)$$

and we want to learn:

$$p(y|x) \quad (3)$$

or, in the paired case used in the training description:

$$(x, y) \quad (4)$$

where x and y correspond to each other.

Examples of image-to-image translation tasks:

- sketch $\rightarrow$ photo
- horse $\rightarrow$ zebra
- summer $\rightarrow$ winter
- segmentation map $\rightarrow$ realistic image
- low-quality $\rightarrow$ high-quality
- edge map $\rightarrow$ image

2 The Traditional Approach: Conditional Generation

The first important idea in this paper is that most diffusion-based image translation methods treat translation as conditional image generation.

Consider a normal DDPM. It does:

$$x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \cdots \rightarrow x_T \quad (5)$$

where:

$$x_0 = \text{real image} \quad (6)$$

and:

$$x_T \approx \mathcal{N}(0, I) \quad (7)$$

During generation:

$$x_T \rightarrow x_{T-1} \rightarrow \cdots \rightarrow x_0 \quad (8)$$

So:

Gaussian noise

x_T

x_{T-1}

$\cdots$

x_0

generated image

For image-to-image translation, however, we have a source image x . So conditional diffusion says:

source image x

noise $\rightarrow x_T \rightarrow x_{T-1} \rightarrow \cdots \rightarrow x_0$
 $\uparrow$

condition

The network might receive:

$$\epsilon_{\theta}(x_t, x, t) \tag{9}$$

or in LDM:

$$\epsilon_{\theta}(z_t, c, t) \tag{10}$$

where c is some representation of the source image.

3 Why Does the Paper Criticize This Approach?

The paper’s criticism is subtle but extremely important.

The source image x is treated as additional conditional information. The diffusion process itself is still basically:

$$\text{noise} \rightarrow \text{target image} \tag{11}$$

The source image is saying: "While generating the target, please pay attention to me."

But there is no explicit diffusion trajectory connecting the source domain to the target domain.

That’s the conceptual problem.

The authors therefore want:

$$A \longleftrightarrow B \tag{12}$$

to be represented directly by the diffusion process itself.

4 The Central Idea of BBDM

This is the single most important thing to understand.

Instead of:

$$\text{noise} \rightarrow B \tag{13}$$

with A acting as a condition, BBDM constructs a stochastic path:

$$A \rightarrow \text{intermediate states} \rightarrow B \tag{14}$$

using a Brownian Bridge.

Imagine two endpoints:

$$x = \text{source image} \tag{15}$$

and:

$$y = \text{target image} \tag{16}$$

The Brownian Bridge says: Start at x , gradually move toward y , while introducing stochasticity along the way.

So:

$$x_0 = x \tag{17}$$

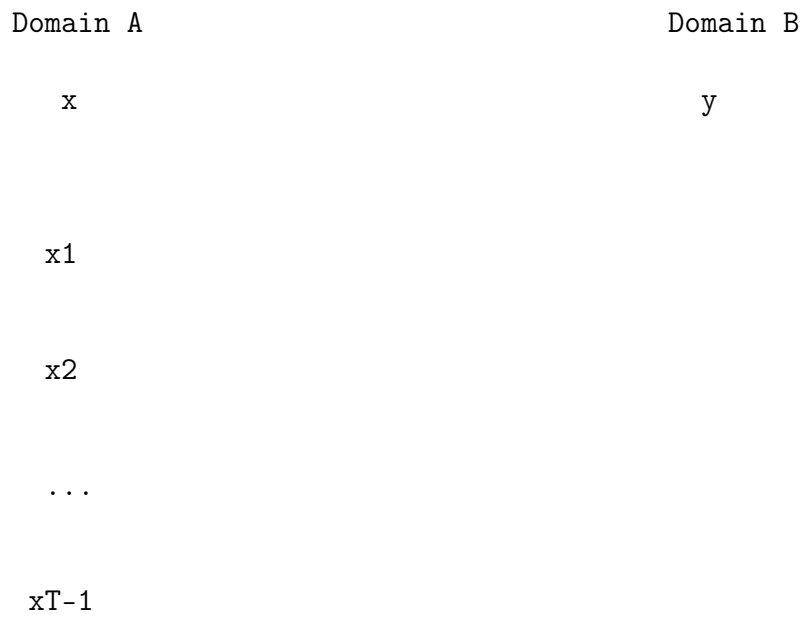
and:

$$x_T = y \tag{18}$$

with intermediate states:

$$x_1, x_2, \dots, x_{T-1} \tag{19}$$

Graphically:



More precisely:

$$x = x_0 \tag{20}$$

and:

$$x_T = y \tag{21}$$

The diffusion process is therefore tied to both endpoints. That's why it is called a bridge.

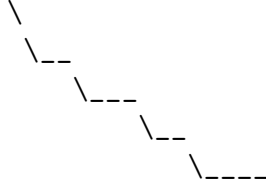
5 Why Brownian Bridge?

Let's understand the ordinary Brownian bridge first.

Suppose you want a random trajectory that:

- starts at x_0
- ends exactly at x_T

A normal Brownian motion might look like:

start


It randomly wanders without being forced to end somewhere specific.
 A Brownian bridge instead says:

x0

 xT

The trajectory is random in the middle but constrained at the endpoints.
 Mathematically:

$$p(x_t|x_0, x_T) = \mathcal{N}\left(\left(1 - \frac{t}{T}\right)x_0 + \frac{t}{T}x_T, \frac{t(T-t)}{T}I\right) \quad (22)$$

The mean is:

$$\mu_t = \left(1 - \frac{t}{T}\right)x_0 + \frac{t}{T}x_T \quad (23)$$

Notice what happens:
 At $t = 0$:

$$\mu_0 = x_0 \quad (24)$$

At $t = T$:

$$\mu_T = x_T \quad (25)$$

In the middle, the mean is approximately an interpolation between the two.

6 Very Important: BBDM is Not Simply Linear Interpolation

This is a common misunderstanding.
 You might think BBDM does:

$$x_t = (1 - m_t)x + m_ty \quad (26)$$

and therefore it simply interpolates from source to target.
 No. It adds Gaussian noise:

$$x_t = (1 - m_t)x + m_ty + \sqrt{\delta_t}\epsilon \quad (27)$$

where:

$$\epsilon \sim \mathcal{N}(0, I) \quad (28)$$

Therefore:

$$\boxed{x_t = (1 - m_t)x + m_ty + \sqrt{\delta_t}\epsilon} \quad (29)$$

This is a stochastic path, not a deterministic straight line.

7 BBDM Works in Latent Space

The paper says: "To speed up the training and inference process, we conduct the diffusion process in the same latent space as used in LDM."

This is exactly the idea from Latent Diffusion Models.

Instead of:

$$I_A \rightarrow \text{diffusion directly in pixels} \quad (30)$$

we first encode:

$$I_A \xrightarrow{E} L_A \quad (31)$$

and:

$$I_B \xrightarrow{E} L_B \quad (32)$$

Then BBDM operates on:

$$L_A, L_B \quad (33)$$

The VQGAN decoder finally reconstructs the image:

$$L_{A \rightarrow B} \xrightarrow{D} I_{A \rightarrow B} \quad (34)$$

So the complete pipeline is:

Source image I_A

VQGAN Encoder

latent x

BBDM

latent translated representation

VQGAN Decoder

Target image $\text{IA} \rightarrow \text{B}$

This is similar to LDM computationally, but the way conditioning/translation is formulated is fundamentally different.

8 DDPM versus BBDM

This distinction is absolutely crucial.

8.1 DDPM

Forward:

$$x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_T \quad (35)$$

where:

$$x_T \sim \mathcal{N}(0, I) \quad (36)$$

Reverse:

$$x_T \rightarrow x_{T-1} \rightarrow \cdots \rightarrow x_0 \quad (37)$$

So:

DATA NOISE

reverse

IMAGE

8.2 BBDM

Forward:

$$x_0 = x \rightarrow x_1 \rightarrow \cdots \rightarrow x_T = y \quad (38)$$

Reverse:

$$x_T = y \rightarrow x_{T-1} \rightarrow \cdots \rightarrow x_0 = x \quad (39)$$

So:

SOURCE x

x_1

x2

...

TARGET y

and the reverse process starts from:

TARGET y

xT-1

xT-2

...

SOURCE x

This is why the paper calls it a bidirectional diffusion process.

9 But Wait — Why Does Inference Start from y ?

This is one of the most confusing points.

During training, we have paired data:

$$(x, y) \tag{40}$$

The forward process is:

$$x_0 = x \tag{41}$$

and:

$$x_T = y \tag{42}$$

Therefore, training teaches the network how to traverse the bridge.

But at inference, if our task is $A \rightarrow B$, we have only the source x , not the actual target y .

So how can sampling start from y ?

The answer is that the reverse process is directionally formulated with the target-domain sample as the starting endpoint during the learned generative process. In the BBDM formulation, the target-domain endpoint is treated as the reference/conditional input and the model learns the reverse bridge.

This is also why the paper's notation can feel counterintuitive compared with ordinary "source $\rightarrow$ target" translation.

The important conceptual point is: The target-domain reference is placed at the diffusion endpoint rather than repeatedly injected as a condition into U-Net.

The exact interpretation depends on which translation direction and training setup is being discussed.

10 Equation (3): Brownian Bridge

The paper gives:

$$p(x_t|x_0, x_T) = \mathcal{N}\left((1 - \frac{t}{T})x_0 + \frac{t}{T}x_T, \frac{t(T-t)}{T}I\right) \quad (43)$$

Let's break this apart.

10.1 Mean

$$(1 - \frac{t}{T})x_0 + \frac{t}{T}x_T \quad (44)$$

Define:

$$m_t = \frac{t}{T} \quad (45)$$

Then:

$$\mu_t = (1 - m_t)x_0 + m_tx_T \quad (46)$$

So:

$t = 0$:

$$\mu_0 = x_0 \quad (47)$$

and:

$t = T$:

$$\mu_T = x_T \quad (48)$$

10.2 Variance

The variance is zero at both endpoints.

Variance:

$$\frac{t(T-t)}{T} \quad (49)$$

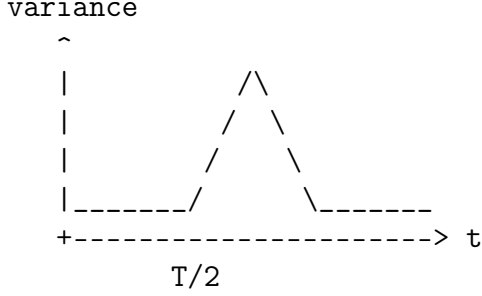
At $t = 0$:

$$\frac{0(T-0)}{T} = 0 \quad (50)$$

At $t = T$:

$$\frac{T(T - T)}{T} = 0 \quad (51)$$

Therefore:



The process is:

- deterministic at x_0
- most stochastic in the middle
- deterministic at x_T

That's the essence of the bridge.

11 Why Don't They Directly Use This Variance?

This is another very important detail.

The original Brownian Bridge has:

$$\delta_t = \frac{t(T - t)}{T} \quad (52)$$

At $t = \frac{T}{2}$, we get:

$$\delta_{T/2} = \frac{T}{4} \quad (53)$$

If T is large, this becomes huge.

For example, $T = 1000$ then:

$$\delta_{500} = 250 \quad (54)$$

That's enormous.

The authors say this makes BBDM difficult to train. So they design a different variance schedule.

12 Their Variance Schedule

They assume approximately:

$$x_0, y \sim \mathcal{N}(0, I) \quad (55)$$

They want the variance of intermediate states to remain controlled.

They define:

$$\delta_t = 1 - (1 - m_t)^2 - m_t^2 \quad (56)$$

This simplifies to:

$$\delta_t = 2(m_t - m_t^2) \quad (57)$$

or:

$$\boxed{\delta_t = 2m_t(1 - m_t)} \quad (58)$$

Then they introduce a scale s :

$$\boxed{\delta_t = 2s(m_t - m_t^2)} \quad (59)$$

13 What Does s Do?

At $m_t = \frac{1}{2}$ we have:

$$\delta_t = 2s \left(\frac{1}{2} - \frac{1}{4} \right) \quad (60)$$

so:

$$\delta_{\max} = \frac{s}{2} \quad (61)$$

Therefore if $s = 1$ then:

$$\delta_{\max} = 0.5 \quad (62)$$

The standard deviation is:

$$\sqrt{0.5} \approx 0.707 \quad (63)$$

So s controls the amount of randomness.

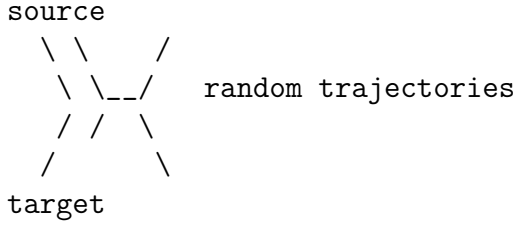
13.1 Small s

Less randomness:

```
source  target
      almost deterministic
```


13.2 Large s

More randomness:



This is why the paper says: sampling diversity can be tuned by the maximum variance.

14 Equation (4): BBDM Forward Marginal

The paper defines:

$$\boxed{q_{BB}(x_t|x_0, y) = \mathcal{N}(x_t; (1 - m_t)x_0 + m_ty, \delta_t I)} \quad (64)$$

This means:

$$\boxed{x_t = (1 - m_t)x_0 + m_ty + \sqrt{\delta_t}\epsilon} \quad (65)$$

where:

$$\epsilon \sim \mathcal{N}(0, I) \quad (66)$$

This equation is the heart of BBDM.

Compare with DDPM.

DDPM:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon \quad (67)$$

BBDM:

$$x_t = (1 - m_t)x_0 + m_ty + \sqrt{\delta_t}\epsilon \quad (68)$$

The difference is enormous.

14.1 DDPM

The endpoint is:

$$\text{Gaussian noise} \quad (69)$$

14.2 BBDM

The endpoint is:

$$\boxed{y} \quad (70)$$

the other image/domain.

15 Let's Use a Simple Numerical Example

Suppose $T = 100$.

At $t = 0$ we have:

$$m_0 = 0 \tag{71}$$

Therefore:

$$x_0 = x \tag{72}$$

At $t = 50$ we have:

$$m_{50} = 0.5 \tag{73}$$

Therefore:

$$x_{50} = 0.5x + 0.5y + \sqrt{0.5}\epsilon \tag{74}$$

So the middle state is roughly:

$$50\% \text{ source} + 50\% \text{ target} + \text{noise} \tag{75}$$

At $t = 100$ we have:

$$m_{100} = 1 \tag{76}$$

Therefore:

$$x_{100} = y \tag{77}$$

So:

t=0	t=50	t=100
x	noisy mixture x + y + noise	y

SOURCE STOCHASTIC BRIDGE TARGET

16 Why Does BBDM Need Equation (8)?

Equation (4) gives:

$$q_{BB}(x_t|x_0, y) \tag{78}$$

This is a marginal distribution. It tells us: Given x_0 and y , what is the distribution of x_t ?

But for a diffusion Markov process, training and sampling require a transition distribution:

$$q(x_t|x_{t-1}, y) \tag{79}$$

That asks: Given the previous state x_{t-1} , what is the distribution of the next state x_t ?

That's why they derive Equation (8).

17 Equation (6)

They write:

$$x_t = (1 - m_t)x_0 + m_ty + \sqrt{\delta_t}\epsilon_t \quad (80)$$

This is simply the reparameterized version of Equation (4).
Similarly:

$$x_{t-1} = (1 - m_{t-1})x_0 + m_{t-1}y + \sqrt{\delta_{t-1}}\epsilon_{t-1} \quad (81)$$

The problem is:

$$q(x_t|x_{t-1}, y) \quad (82)$$

contains x_0 , which we don't directly know from x_{t-1} .
So they algebraically eliminate x_0 .

18 Equation (8)

They obtain:

$$q_{BB}(x_t|x_{t-1}, y) = \mathcal{N}\left(x_t; \frac{1 - m_t}{1 - m_{t-1}}x_{t-1} + \left(m_t - \frac{1 - m_t}{1 - m_{t-1}}m_{t-1}\right)y, \delta_{t|t-1}I\right) \quad (83)$$

Don't try to memorize this entire equation. Understand its structure:

$$x_t = a_tx_{t-1} + b_ty + \text{noise} \quad (84)$$

where a_t and b_t are determined by the bridge schedule.

This tells us: The next state depends on the previous state, the destination y , and some stochasticity.

19 What Happens at $t = T$?

At the destination:

$$m_T = 1 \quad (85)$$

Therefore:

$$x_T = y \quad (86)$$

The variance becomes:

$$\delta_T = 0 \tag{87}$$

So:

$$\boxed{x_T = y} \tag{88}$$

with probability 1.

This is very different from DDPM.

DDPM:

$$x_T \sim \mathcal{N}(0, I) \tag{89}$$

BBDM:

$$x_T = y \tag{90}$$

20 Reverse Process

Now comes the most important part for generation.

In DDPM:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(\mu_\theta(x_t, t), \tilde{\beta}_t I) \tag{91}$$

The network predicts how to move:

$$x_t \rightarrow x_{t-1} \tag{92}$$

BBDM does:

$$p_\theta(x_{t-1}|x_t, y) = \mathcal{N}(\mu_\theta(x_t, t), \tilde{\delta}_t I) \tag{93}$$

So mathematically, y appears in the reverse distribution.

But here's the subtlety: The authors emphasize that y is not repeatedly injected into the neural network as a separate condition, unlike conditional LDMs.

Instead:

$$x_T = y \tag{94}$$

So the target-domain image establishes the starting point of the reverse diffusion.

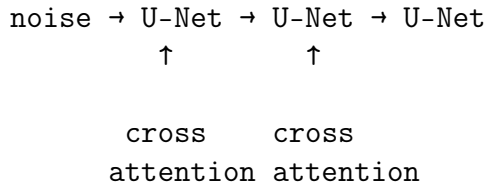
21 This is the Key Difference from LDM

Suppose conventional conditional LDM does:

source image

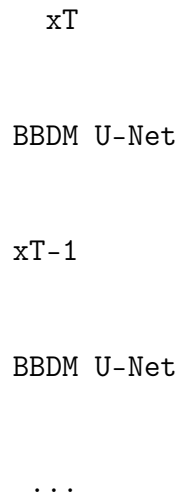
encoder

condition c



The source condition is injected at multiple layers/timesteps.
BBDM instead conceptually does:

target/reference endpoint y



The source-target relationship is encoded by the bridge process.

That is what the authors mean by: "directly builds the mapping between the input and output domains."

22 Why Do They Say "Bidirectional"?

Because the Brownian Bridge is defined by two endpoints.

Ordinary diffusion essentially learns:

$$\text{data} \leftrightarrow \text{noise} \tag{95}$$

BBDM learns:

$$\boxed{A \leftrightarrow B} \tag{96}$$

with the diffusion trajectory connecting them.

You can visualize:

stochastic bridge

Domain A
x

Domain B
y

The process isn't just:

$$A \rightarrow \text{noise} \rightarrow B \quad (97)$$

Instead, the two domains are the endpoints of the stochastic process.

23 Training Objective

Now we get to the difficult mathematical part.

They start with the ELBO:

$$\text{ELBO} = -\mathbb{E}_q \left[D_{KL}(q_{BB}(x_T|x_0, y)|p(x_T|y)) + \sum_{t=2}^T D_{KL}(q_{BB}(x_{t-1}|x_t, x_0, y)|p_\theta(x_{t-1}|x_t, y)) - \log p_\theta(x_0|x_1, y) \right] \quad (98)$$

At first glance this looks terrifying. But conceptually it is the same idea as DDPM.

The true reverse distribution is:

$$q(x_{t-1}|x_t, x_0, y) \quad (99)$$

The model learns:

$$p_\theta(x_{t-1}|x_t, y) \quad (100)$$

Training tries to make them close:

$$\boxed{q \approx p_\theta} \quad (101)$$

using KL divergence.

24 Why Does the First KL Term Disappear?

They say: Since x_T is equal to y in Brownian Bridge, the first term can be seen as a constant and ignored.

Because:

$$x_T = y \quad (102)$$

deterministically.

There isn't a learned Gaussian noise endpoint like DDPM. So the model doesn't need to learn:

$$q(x_T|x_0) \rightarrow p(x_T) \quad (103)$$

The endpoint is already specified.

25 The Posterior $q(x_{t-1}|x_t, x_0, y)$

The authors derive:

$$q_{BB}(x_{t-1}|x_t, x_0, y) = \mathcal{N}\left(x_{t-1}; \tilde{\mu}_t(x_t, x_0, y), \tilde{\delta}_t I\right) \quad (104)$$

This is the equivalent of the DDPM posterior:

$$q(x_{t-1}|x_t, x_0) \quad (105)$$

But BBDM has an additional endpoint y .

Therefore:

$$\boxed{q(x_{t-1}|x_t, x_0, y)} \quad (106)$$

rather than:

$$q(x_{t-1}|x_t, x_0) \quad (107)$$

26 Why is x_0 a Problem?

During training, x_0 is known.

But during inference, we don't know x_0 . That's exactly the same problem DDPM has.

In DDPM:

- Training: x_0 known
- Inference: x_0 unknown

So the network predicts noise instead of directly predicting x_0 or the exact posterior mean.

BBDM uses the same trick.

27 Reparameterization

From Equation (4):

$$x_t = (1 - m_t)x_0 + m_ty + \sqrt{\delta_t}\epsilon \quad (108)$$

Rearrange conceptually:

$$x_0 = \frac{x_t - m_ty - \sqrt{\delta_t}\epsilon}{1 - m_t} \quad (109)$$

The authors substitute this into the posterior mean. This eliminates explicit dependence on x_0 .

Then the posterior mean can be expressed as:

$$\tilde{\mu}_t(x_t, y) = c_{xt}x_t + c_{yt}y + c_{\epsilon t}\epsilon \quad (110)$$

This is extremely important.

The reverse mean depends on:

- current latent x_t
- endpoint y
- noise ϵ

28 But the Network Doesn't Know the True Noise

Exactly.

During training we generated:

$$x_t = (1 - m_t)x_0 + m_ty + \sqrt{\delta_t}\epsilon \quad (111)$$

So we know the actual ϵ .

The neural network learns:

$$\epsilon_\theta(x_t, t) \quad (112)$$

to predict it.

Therefore:

$$\epsilon \approx \epsilon_\theta(x_t, t) \quad (113)$$

This is analogous to ordinary DDPM.

29 The Surprising Part: What is the Network Predicting?

The paper's training objective is:

$$\mathbb{E} \left[c_{\epsilon t} \left| m_t(y - x_0) + \sqrt{\delta_t}\epsilon - \epsilon_\theta(x_t, t) \right|^2 \right] \quad (114)$$

Notice that the target is:

$$m_t(y - x_0) + \sqrt{\delta_t}\epsilon \quad (115)$$

not simply ϵ .

This is one of the most important differences from standard DDPM.

30 Why is the Target Not Just ϵ ?

Look at the bridge equation:

$$x_t = (1 - m_t)x_0 + m_ty + \sqrt{\delta_t}\epsilon \quad (116)$$

Rearrange:

$$x_t - x_0 = m_t(y - x_0) + \sqrt{\delta_t}\epsilon \quad (117)$$

Therefore:

$$\boxed{x_t - x_0 = m_t(y - x_0) + \sqrt{\delta_t}\epsilon} \quad (118)$$

The training target is essentially the total displacement away from the source, consisting of:

- Domain-translation component: $m_t(y - x_0)$
- plus stochastic component: $\sqrt{\delta_t}\epsilon$

So BBDM's network is learning not just "remove random noise," but the effective transformation associated with moving from one endpoint toward the other.

That's a beautiful way to understand the method.

31 Compare DDPM and BBDM Prediction Targets

31.1 DDPM

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon \quad (119)$$

Network predicts:

$$\epsilon_\theta(x_t, t) \approx \epsilon \quad (120)$$

31.2 BBDM

$$x_t = (1 - m_t)x_0 + m_ty + \sqrt{\delta_t}\epsilon \quad (121)$$

Network predicts a quantity related to:

$$m_t(y - x_0) + \sqrt{\delta_t}\epsilon \quad (122)$$

So:

DDPM:

current state

↓

estimate random noise

BBDM:

current state
 $\downarrow$
estimate bridge displacement/noise
 $\downarrow$
move along A B bridge

32 Equation (14)

They write:

$$\boxed{\mu_{\theta}(x_t, y, t) = c_{xt}x_t + c_{yt}y + c_{\epsilon t}\epsilon_{\theta}(x_t, t)} \quad (123)$$

This is extremely useful to understand.

The predicted reverse mean has three pieces:

$$\boxed{\text{reverse mean} = \text{current state contribution} + \text{destination contribution} + \text{network prediction}} \quad (124)$$

Specifically:

- $c_{xt}x_t$ says: Where am I now?
- $c_{yt}y$ says: Where is the endpoint?
- $c_{\epsilon t}\epsilon_{\theta}$ says: What did the learned model say about the bridge/noise transformation?

33 Sampling Algorithm

Now let's understand Algorithm 2.

They write:

1. sample conditional input $x_T = y$
2. for $t = T, \dots, 1$:
3. $z \sim \mathcal{N}(0, I)$, if $t > 1$
4. $x_{t-1} = c_{xt}x_t + c_{yt}y - c_{\epsilon t}\epsilon_{\theta}(x_t, t) + \sqrt{\tilde{\delta}_t}z$
5. return x_0

Let's translate this into plain English.

34 Step 1: Initialize

They set:

$$x_T = y \quad (125)$$

This is fundamentally different from DDPM.

DDPM:

$$x_T \sim \mathcal{N}(0, I) \quad (126)$$

BBDM:

$$\boxed{x_T = y} \quad (127)$$

So the reverse process begins at the target/reference endpoint.

35 Step 2: Run Backwards

They iterate:

$$T, T-1, T-2, \dots, 1 \quad (128)$$

At each step:

$$x_t \rightarrow x_{t-1} \quad (129)$$

So:

x_T
 $\downarrow$
 x_{T-1}
 $\downarrow$
 x_{T-2}
 $\downarrow$
 $\dots$
 $\downarrow$
 x_1
 $\downarrow$
 x_0

36 Step 3: Add Stochasticity

They sample:

$$z \sim \mathcal{N}(0, I) \quad (130)$$

when $t > 1$.

At the final step $t = 1$, they set $z = 0$.

Why? Because we want the final output x_0 to be clean rather than injecting additional noise at the very end.

37 Step 4: Calculate x_{t-1}

The equation is:

$$x_{t-1} = c_{xt}x_t + c_{yt}y - c_{\epsilon t}\epsilon_{\theta}(x_t, t) + \sqrt{\tilde{\delta}_t}z \quad (131)$$

Think of it as:

next state = current state + endpoint information – predicted transformation + randomness

(132)

This is the reverse bridge step.

38 What Happens Throughout Sampling?

Conceptually:

reverse Brownian bridge

xT = y

xT-1

xT-2

...

x2

x1

x0

translated latent/image

At each step, the neural network predicts how to move toward the other endpoint.

39 One Important Conceptual Issue About the Notation

The paper uses $x_0 = x$ and $x_T = y$ for the forward bridge.

Then Algorithm 2 starts with:

$$x_T = y \tag{133}$$

Therefore the reverse trajectory is:

$$y \rightarrow x \tag{134}$$

This means the mathematical process can be viewed as learning one direction of the bridge, and the opposite translation direction can be obtained by swapping the roles of the two domains.

So:

- $A \rightarrow B$ could be represented by choosing $x_0 = A, x_T = B$
- $B \rightarrow A$ by swapping $x_0 = B, x_T = A$

This is one reason the authors emphasize the bidirectional nature of the framework.

40 Why Doesn't BBDM Use Cross-Attention Like LDM?

This connects directly to the LDM discussion.

In conditional LDM:

text/source condition

embedding

cross-attention

U-Net

The condition is repeatedly provided to the U-Net.

For example:

$$Q = W_Q h \tag{135}$$

from image latent features, while:

$$K = W_K c, \quad V = W_V c \tag{136}$$

come from the conditioning representation.

Then:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V \tag{137}$$

BBDM's central idea is different.

It doesn't need the source/target relation to be represented as an external condition at every U-Net layer.

Instead:

$$\boxed{\text{the diffusion trajectory itself represents the domain relation}} \quad (138)$$

The endpoint y defines the bridge.

41 This Does NOT Mean BBDM Has No U-Net

It still uses a neural network architecture based on the diffusion paradigm.

The important difference is what the U-Net is asked to model.

Conditional LDM:

$$\epsilon_{\theta}(z_t, c, t) \quad (139)$$

where c is an explicit condition.

BBDM:

$$\epsilon_{\theta}(x_t, t) \quad (140)$$

with the bridge endpoint incorporated into the diffusion formulation.

So the distinction is:

Table 1: Conditional LDM vs BBDM

Conditional LDM	BBDM
Noise $\rightarrow$ target	One domain $\leftrightarrow$ another domain
Source acts as condition	Endpoint defines bridge
Condition injected into network	Bridge encodes domain relation
Conditional generation	Direct translation process
Cross-attention often used	No need for source conditioning through cross-attention as the central mechanism
Target distribution guided by condition	Translation modeled through stochastic bridge

42 Why Does This Potentially Help with Domain Gaps?

Imagine:

$$A = \text{sketches} \quad (141)$$

and:

$$B = \text{photographs} \quad (142)$$

These are very different distributions.

A conditional generation approach effectively says: Generate a photograph, while looking at this sketch.

But BBDM says: Learn a stochastic process whose endpoints are sketch and photograph representations.

Therefore the model explicitly learns the transition:

$$P_A \leftrightarrow P_B \tag{143}$$

The domain gap becomes part of the diffusion process itself.

43 Why Does the Paper Criticize Inpainting/Super-Resolution?

They say conditional diffusion works particularly well when the condition and target have high similarity.

For example, super-resolution:

low-res image

same scene, higher detail

The condition and output share huge amounts of information.

Similarly, inpainting:

known image + mask

same image with missing region filled

Again, the condition is very close to the target.

But consider:

- sketch $\rightarrow$ photograph
- summer $\rightarrow$ winter

The relationship is much more complicated.

The paper argues that conventional conditional diffusion has less explicit theoretical structure for such domain translation.

44 Why GANs are Criticized

The paper discusses Pix2Pix and other GAN methods.

GANs can generate very realistic images, but two issues arise.

44.1 Training instability

GAN training involves two networks: G and D .

Generator:

$$G(x) \rightarrow y \quad (144)$$

Discriminator:

$$D(y) \quad (145)$$

They compete. This can be unstable.

44.2 Mode collapse

Suppose there are many valid target images:

$$y_1, y_2, y_3, \dots \quad (146)$$

A GAN might produce only y_2 again and again.

So:

Possible outputs:

A B1

A B2

A B3

A B4

GAN may learn:

A B2

A B2

A B2

A B2

Diffusion models are generally much better at representing multimodal distributions.

45 Why BBDM Can Produce Diverse Outputs

Remember:

$$x_t = (1 - m_t)x_0 + m_ty + \sqrt{\delta_t}\epsilon \quad (147)$$

There is randomness:

$$\epsilon \sim \mathcal{N}(0, I) \quad (148)$$

And during reverse sampling:

$$z \sim \mathcal{N}(0, I) \quad (149)$$

Therefore different random trajectories can produce different outputs.

The parameter s controls this diversity.

$$\delta_t = 2s(m_t - m_t^2) \quad (150)$$

Higher s :

$$\Rightarrow \text{more stochasticity} \Rightarrow \text{more diverse samples} \quad (151)$$

Lower s :

$$\Rightarrow \text{less stochasticity} \Rightarrow \text{more deterministic translation} \quad (152)$$

46 The Most Important Equations to Understand

If you're studying this paper deeply, don't try to memorize every equation. Focus on these.

46.1 Equation 4

$$q_{BB}(x_t|x_0, y) = \mathcal{N}((1 - m_t)x_0 + m_ty, \delta_t I) \quad (153)$$

This defines the forward bridge.

46.2 Equation 5

$$\delta_t = 2s(m_t - m_t^2) \quad (154)$$

This controls stochasticity.

46.3 Equation 6

$$x_t = (1 - m_t)x_0 + m_ty + \sqrt{\delta_t}\epsilon \quad (155)$$

This is the practical sampling equation for creating noisy bridge states.

46.4 Equation 9

$$p_\theta(x_{t-1}|x_t, y) = \mathcal{N}(\mu_\theta(x_t, t), \tilde{\delta}_t I) \quad (156)$$

This defines the learned reverse process.

46.5 Equation 14

$$\mu_\theta = c_{xt}x_t + c_{yt}y + c_{\epsilon t}\epsilon_\theta \quad (157)$$

This tells us how the network prediction is converted into the reverse mean.

47 The Entire Training Process in Simple Terms

Suppose we have:

$$(x, y) \tag{158}$$

where:

$$x = \text{domain A image} \tag{159}$$

and:

$$y = \text{domain B image} \tag{160}$$

47.1 Step 1

Encode them into latent space:

$$x = E(I_A) \tag{161}$$

$$y = E(I_B) \tag{162}$$

47.2 Step 2

Randomly choose:

$$t \sim U(1, T) \tag{163}$$

47.3 Step 3

Calculate:

$$m_t = \frac{t}{T} \tag{164}$$

47.4 Step 4

Calculate:

$$\delta_t = 2s(m_t - m_t^2) \tag{165}$$

47.5 Step 5

Sample Gaussian noise:

$$\epsilon \sim \mathcal{N}(0, I) \tag{166}$$

47.6 Step 6

Create a bridge state:

$$\boxed{x_t = (1 - m_t)x + m_ty + \sqrt{\delta_t}\epsilon} \quad (167)$$

47.7 Step 7

Give the network:

$$(x_t, t) \quad (168)$$

The network predicts:

$$\epsilon_\theta(x_t, t) \quad (169)$$

47.8 Step 8

Compare prediction with the appropriate bridge target:

$$m_t(y - x) + \sqrt{\delta_t}\epsilon \quad (170)$$

47.9 Step 9

Backpropagate.

That's it.

48 Training Algorithm in One Picture

paired images

Image A

Image B

VQGAN Encoder

VQGAN Encoder

x

y

choose t

calculate m_t , t

sample $\sim N(0, I)$

$x_t = (1 - m_t)x + m_t y + \epsilon$

U-Net / network
(x_t, t)

loss

backpropagation

49 Sampling Algorithm in One Picture

Conceptually:

endpoint/reference
 y

x_T

Network
(x_t, t)

x_{T-1}

Network
(x_t, t)

x_{T-2}

...

x_1

x0

VQGAN Decoder

translated image

50 Accelerated Sampling

The authors then discuss DDIM-style acceleration.

Normal diffusion might require:

$$T = 1000 \tag{171}$$

steps. That's expensive.

Instead of:

$$1000 \rightarrow 999 \rightarrow 998 \rightarrow \dots \rightarrow 1 \tag{172}$$

we can select only some timesteps:

$$\tau_1, \tau_2, \dots, \tau_S \tag{173}$$

For example:

$$1000, 995, 990, \dots, 5, 1 \tag{174}$$

Or a more aggressively spaced sequence.

Then:

$$S \ll T \tag{175}$$

The paper chooses:

$$\boxed{S = 200} \tag{176}$$

as the default.

51 Why Can They Skip Timesteps?

Because they construct a non-Markovian inference process that preserves the desired marginal distributions.

This is the same general idea behind DDIM.

Instead of simulating every tiny diffusion step:

x1000
 $\downarrow$
x999
 $\downarrow$
x998
 $\downarrow$
...

we can do:

x1000
 $\downarrow$
x950
 $\downarrow$
x900
 $\downarrow$
...
 $\downarrow$
x1

while maintaining the appropriate distributional behavior.
This greatly reduces inference cost.

52 The Role of $S = 200$

If the original process has $T = 1000$ and inference uses $S = 200$, then approximately only 20% as many reverse steps are performed.

So there is a tradeoff:

$$\text{more steps} \Rightarrow \text{better accuracy but slower} \quad (177)$$

$$\text{fewer steps} \Rightarrow \text{faster but potentially lower quality} \quad (178)$$

53 The Biggest Conceptual Difference: DDPM vs Conditional Diffusion vs BBDM

This is what I recommend you remember for your research.

53.1 Standard DDPM

$$\text{Forward: } \boxed{\text{Noise} \rightarrow \text{Image}} \quad (179)$$

$$\text{Image} \rightarrow \text{Noise} \quad (180)$$

$$\text{Reverse: } \text{Noise} \rightarrow \text{Image} \quad (181)$$

53.2 Conditional DDPM/LDM

$$\boxed{\text{Noise} + \text{Condition} \rightarrow \text{Target Image}} \quad (182)$$

The condition is supplied to the denoising network.

53.3 BBDM

$$\boxed{\text{Domain A} \leftrightarrow \text{Domain B}} \quad (183)$$

The diffusion process itself is constructed as a stochastic bridge between the two domains.

That is the core innovation.

54 A Very Useful Mental Model

Think of three people walking.

54.1 DDPM

Person A: "I am at random noise. Please guide me toward an image."

NOISE IMAGE

54.2 Conditional LDM

Person A: "I am at random noise. Here is a source image. Please use it to guide me toward the target."

SOURCE

NOISE U-NET TARGET

54.3 BBDM

Person A: "I know both endpoints. Learn the stochastic path connecting them."

SOURCE stochastic bridge TARGET

The path itself becomes the translation model.

55 Why This is Called a "Brownian Bridge Diffusion Model"

Break the name apart:

55.1 Brownian

because stochastic Gaussian perturbations are involved.

55.2 Bridge

because the process is constrained between two endpoints:

$$x_0 = x, \quad x_T = y \quad (184)$$

55.3 Diffusion

because we progressively move through noisy intermediate states and learn the reverse process.

Therefore:

BBDM = stochastic diffusion process connecting two endpoints

 (185)

56 One Subtle Point You Should Be Careful About

The paper sometimes uses language like: "reference image y " and then says $x_T = y$.

This can initially sound like ordinary conditional diffusion.

But don't interpret it as:

$$y \rightarrow \text{condition encoder} \rightarrow \text{cross-attention at every U-Net block} \quad (186)$$

That is not the central BBDM mechanism.

Instead:

y defines the bridge endpoint

 (187)

and the learned process is designed around that endpoint.

This is the conceptual distinction the paper is trying very hard to establish.

57 The Complete Mathematical Story

You can summarize the entire BBDM mathematically as:

Given:

$$x \in A, \quad y \in B \quad (188)$$

Define:

$$m_t = \frac{t}{T} \quad (189)$$

Define variance:

$$\delta_t = 2s(m_t - m_t^2) \quad (190)$$

Forward bridge:

$x_t = (1 - m_t)x + m_ty + \sqrt{\delta_t}\epsilon$

 (191)

with:

$$\epsilon \sim \mathcal{N}(0, I) \quad (192)$$

Therefore:

$$x_0 = x \quad (193)$$

and:

$$x_T = y \quad (194)$$

Train network:

$$\epsilon_\theta(x_t, t) \quad (195)$$

to estimate the bridge transformation/noise target.

Reverse mean:

$$\boxed{\mu_\theta = c_{xt}x_t + c_{yt}y - c_{\epsilon t}\epsilon_\theta} \quad (196)$$

with the sign convention appearing in the sampling equation.

Reverse sampling:

$$\boxed{x_{t-1} = c_{xt}x_t + c_{yt}y - c_{\epsilon t}\epsilon_\theta(x_t, t) + \sqrt{\tilde{\delta}_t}z} \quad (197)$$

until x_0 is obtained.

Decode:

$$I_{A \rightarrow B} = D(x_0) \quad (198)$$

58 If You Understand Only 7 Things from This Paper

Remember these:

1. Image-to-image translation $A \rightarrow B$ means learning a mapping between two image domains.
2. Conventional diffusion: noise $\rightarrow$ image
3. Conditional diffusion: noise + condition $\rightarrow$ target
4. BBDM changes the formulation: $A \rightarrow$ stochastic bridge $\rightarrow B$ rather than treating A merely as an external condition.
5. The bridge equation $\boxed{x_t = (1 - m_t)x_0 + m_ty + \sqrt{\delta_t}\epsilon}$ is the most important equation.
6. The endpoints are fixed: $\boxed{x_0 = x, \quad x_T = y}$ and the variance is zero at both endpoints.
7. The network learns the reverse bridge: $x_t \rightarrow x_{t-1}$ until the translated latent is obtained.

59 Final Picture

If you keep one diagram in your head, keep this:

BBDM

Domain A
(source)
 x_0

Domain B
(target)
 $y=x_T$

Brownian Bridge diffusion

$x_1 \ x_2 \ \dots \ x_{T-1}$

stochastic intermediate states

Forward process:

$x_0 \ x_1 \ \dots \ x_T=y$

Reverse process:

$x_T=y \ x_{T-1} \ \dots \ x_0$

At each reverse step:

U-Net
 (x_t, t)

predict bridge
transformation

$x_t \rightarrow x_{t-1}$

Unlike conditional LDM:

Condition Cross-Attention U-Net

BBDM:

Endpoint + bridge formulation diffusion trajectory

The deepest idea of the paper is therefore not the complicated coefficients c_{xt}, c_{yt}, c_{et} . Those coefficients are derived so that the reverse process is mathematically consistent with the Brownian-bridge forward process.

The real conceptual contribution is:

$$\boxed{\text{Instead of asking diffusion to generate B conditioned on A,}} \quad (199)$$

$$\boxed{\text{BBDM constructs a stochastic diffusion bridge whose two endpoints are A and B.}} \quad (200)$$

That is why the authors call it direct domain-to-domain translation rather than conditional generation.

60 The Key Problem: Conditional Information, Leverage, and Sampling Steps

These three terms — conditional information, leverage, and sampling steps — are connected, but they mean different things in diffusion models.

60.1 Conditional Information

Suppose your task is:

$$\text{source image } x \rightarrow \text{target image } y \quad (201)$$

A conditional diffusion model doesn't just generate y from noise. It generates y while receiving x .

So the model predicts:

$$\epsilon_{\theta}(x_t, t, \underbrace{x}_{\text{condition}}) \quad (202)$$

Here, x is the conditional information.

For example:

Source image

condition

noisy target $x_t \rightarrow \text{U-Net} \rightarrow \text{predicted noise}$

The source tells the model things like:

- where the object is
- its shape
- its spatial arrangement
- its identity
- structural details

But the diffusion process is still happening on x_t , which is the noisy target.

60.2 What Does "Leverage" Mean?

When a paper says something like: "leverage conditional information", it simply means: Use the information available in the condition to guide the generation process.

For example:

$$x_{\text{source}} = \text{sketch} \quad (203)$$

and:

$$y = \text{realistic image} \quad (204)$$

The model has random noise z_T and wants to generate y .

The source sketch provides useful information:

random noise

denoising

source sketch

So the model leverages the sketch to decide what the generated image should look like.

It does not mean that the source image replaces the noise.

60.3 Where Does the Conditional Information Enter?

For an LDM/Stable-Diffusion-style model:

Source image

Vision encoder

condition embeddings

U-Net

noisy latent z_t

+

condition information

predicted noise

The exact mechanism can be cross-attention, concatenation, ControlNet-style injection, etc.

The important point is:

$$\boxed{\text{condition guides the denoising}} \quad (205)$$

60.4 Now: What are Sampling Steps?

This is a completely different concept.

During training, we might have:

$$x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \cdots \rightarrow x_T \quad (206)$$

But during generation, we start with noise:

$$x_T \quad (207)$$

and repeatedly denoise:

$$x_T \rightarrow x_{T-1} \rightarrow x_{T-2} \rightarrow \cdots \rightarrow x_0 \quad (208)$$

Each transition is a sampling/denoising step.

For example, with 5 sampling steps:

noise

step 1

step 2

step 3

step 4

step 5

generated image

With 50 steps:

noise

step 1

step 2

step 3
 ...
 step 49
 step 50
 ↓
 image

60.5 Why Do We Need Many Sampling Steps?

Because diffusion doesn't normally transform:

$$\text{noise} \rightarrow \text{image} \quad (209)$$

in one jump.

Instead, it learns a sequence of small denoising transformations.

At each step:

$$x_t \rightarrow x_{t-1} \quad (210)$$

The neural network predicts how to remove some of the noise.

So:

$$\boxed{\text{many small corrections} \rightarrow \text{final image}} \quad (211)$$

60.6 Put Everything Together

Now you can see the complete conditional diffusion process.

60.6.1 Training

We take target image y :

$$y_0 \rightarrow y_t \quad (212)$$

by adding noise.

At the same time, we provide the source x :

$$\boxed{\epsilon_\theta(y_t, t, x)} \quad (213)$$

The network learns: Given this noisy target, this timestep, and this source information, what noise should I remove?

60.6.2 Sampling

We start with noise:

$$y_T \sim \mathcal{N}(0, I) \quad (214)$$

Then repeatedly:

$$y_T \rightarrow y_{T-1} \rightarrow \cdots \rightarrow y_0 \quad (215)$$

At every sampling step, the source condition can be supplied:

source x

noise $\rightarrow$ U-Net $\rightarrow$ less noisy $\rightarrow$ U-Net $\rightarrow$ less noisy $\rightarrow \dots \rightarrow$ target
 $\uparrow$ $\uparrow$
condition condition

Therefore:

$$\boxed{\text{conditional information} = \text{what guides generation}} \quad (216)$$

$$\boxed{\text{leverage} = \text{use that information effectively}} \quad (217)$$

$$\boxed{\text{sampling steps} = \text{number of denoising transitions used to generate the image}} \quad (218)$$

61 The Criticism in the BBDM Paper

The criticism in the BBDM paper is essentially:

61.1 Conditional diffusion

$$\boxed{\text{noise} \rightarrow \text{target} \quad \text{while leveraging source as condition}} \quad (219)$$

The source is outside the diffusion trajectory.

61.2 Brownian-bridge diffusion

$$\boxed{\text{source} \rightarrow \text{intermediate states} \rightarrow \text{target}} \quad (220)$$

The source and target are both endpoints of the diffusion trajectory.
That's the conceptual difference.

62 The Key Distinction: Source as Condition vs Source as Trajectory

The key problem is not that conditioning on the source image is wrong. The problem is how the source image is used.

Suppose the task is:

$$x_{\text{source}} \rightarrow y_{\text{target}} \quad (221)$$

For example: sketch $\rightarrow$ photo, edge map $\rightarrow$ photograph, summer $\rightarrow$ winter, low-resolution $\rightarrow$ high-resolution.

62.1 What a Conventional Conditional Diffusion Model Does

A typical conditional diffusion model learns something like:

$$p_{\theta}(y_0 \mid x_{\text{source}}) \quad (222)$$

The diffusion process itself is applied only to the target y :

$$y_0 \rightarrow y_1 \rightarrow y_2 \rightarrow \cdots \rightarrow y_T \quad (223)$$

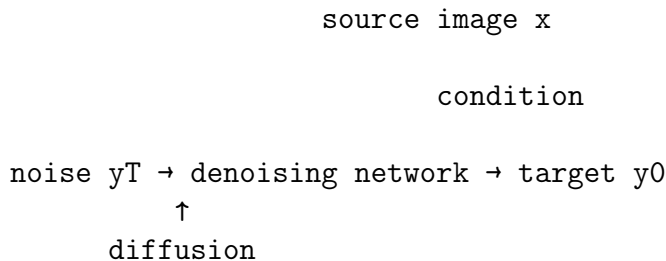
where y_T is approximately Gaussian noise.

During denoising:

$$y_T \rightarrow y_{T-1} \rightarrow \cdots \rightarrow y_0 \quad (224)$$

the model receives x_{source} as an additional condition.

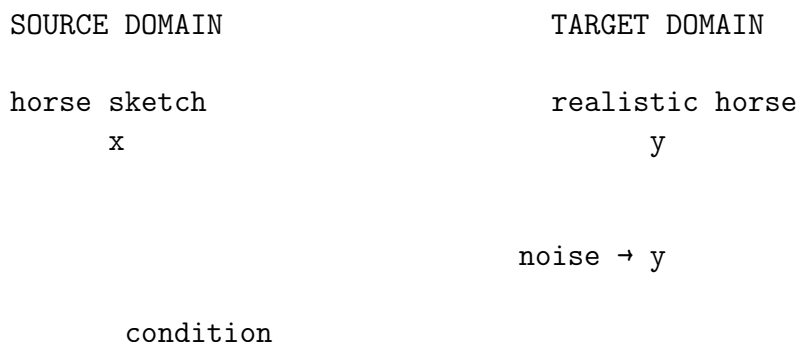
So conceptually:



The source image does not participate in the diffusion trajectory.

62.2 Why Can That Be a Problem?

Imagine:



The model has to learn two things simultaneously:

1. How to generate an image from noise
2. How to use the source image to make that generated image correspond to the source

The diffusion trajectory itself knows nothing about the source. It is fundamentally learning:

$$\text{noise} \rightarrow \text{target image} \quad (225)$$

with the source acting as auxiliary information.

62.3 The Deeper Issue: Domain Gap

This becomes particularly problematic when the source and target domains are very different.

Suppose:

$$x \in \text{sketch domain} \tag{226}$$

and:

$$y \in \text{photo domain} \tag{227}$$

The source contains information such as:

- object structure
- contours
- geometry
- spatial arrangement

while the target contains:

- texture
- color
- lighting
- realistic appearance

There is a substantial domain gap:

$$\text{Sketch} \neq \text{Photo} \tag{228}$$

A conditional diffusion model has to somehow learn the mapping:

$$x_{\text{sketch}} \rightarrow y_{\text{photo}} \tag{229}$$

while its actual generative process is still:

$$\text{noise} \rightarrow y_{\text{photo}} \tag{230}$$

The source information is injected into this process rather than being transformed into the target through a continuous generative path.

62.4 Think of It This Way

Suppose I tell you: "Here is a sketch of a dog. Now draw a realistic dog. Keep looking at this sketch while drawing."

You start from a blank canvas and progressively create the realistic dog. The sketch is guidance.

But you never actually transform:

$$\text{sketch} \rightarrow \text{intermediate representation} \rightarrow \text{realistic dog} \quad (231)$$

Instead:

$$\text{blank/noise} \rightarrow \text{realistic dog} \quad \text{with sketch as guidance} \quad (232)$$

That's essentially the criticism.

62.5 What Would Be More Natural?

For image-to-image translation, you might want the diffusion process itself to represent the transition between the two domains.

Something conceptually like:

source image

x

x

x

...

target domain

Now the intermediate states can gradually move from the source-domain distribution toward the target-domain distribution.

So instead of merely saying: "Generate a target image while looking at the source." we want: "Start from the source and progressively transform it into the target."

That's a much stronger formulation for image-to-image translation.

62.6 Why is This Especially Useful?

Because the source image already contains useful structure.

For example:

source

sketch

intermediate

shape +
rough color

target

realistic
dog photo

A diffusion trajectory connecting the domains can preserve information progressively.

A pure conditional-generation formulation instead asks the network to recover/use that source information while simultaneously generating the target from noise.

62.7 This is the Important Distinction

Conditional diffusion:

$$\boxed{x_{\text{source}} + \text{noise} \rightarrow y_{\text{target}}} \quad (233)$$

The source is conditioning information.

Translation-oriented diffusion:

$$\boxed{x_{\text{source}} \rightarrow \text{intermediate states} \rightarrow y_{\text{target}}} \quad (234)$$

The source is part of the generative trajectory itself.

62.8 One Subtle Point

This does not mean conditional diffusion cannot perform image-to-image translation well. Models such as ControlNet, SDEdit-style approaches, and conditional LDMs can produce excellent results.

The paper’s criticism is more conceptual: If image-to-image translation is fundamentally a transformation between two image domains, simply treating the source as an external condition does not explicitly model the transition between those domains.

That’s why the authors want a diffusion formulation where the source and target are connected by the diffusion process itself, rather than having the source merely tell a noise-to-target generator what to produce.

63 The Intuition: Starting from Source vs Starting from Noise

The important distinction is between:

- using the source image as information/guidance, and
- actually starting the diffusion trajectory from the source image and transforming it into the target.

And yes, the intuition about "converting an already structured image is easier than starting from blank noise" is part of why the latter can be advantageous.

63.1 Forget Domains for a Moment

Suppose we have:

Source image x

Target image y

cartoon cat $\rightarrow$ realistic cat

The goal is simply:

$$x \rightarrow y \tag{235}$$

Now consider two approaches.

63.2 Approach A — Source is Only Conditioning

The diffusion model starts with noise:

source image

information

noise $\rightarrow$ diffusion model $\rightarrow$ target

Mathematically, approximately:

$$\boxed{z_T \rightarrow z_{T-1} \rightarrow \cdots \rightarrow z_0 = y} \quad (236)$$

while the network receives x :

$$\epsilon_{\theta}(z_t, t, \boxed{x}) \quad (237)$$

So the model is essentially saying: "I'm going to create an image from noise. I'll look at the source image to know what image I should create."

63.3 Approach B — Start from the Source

Instead, imagine corrupting the source:

source

slightly noisy source

more noisy

very noisy

...

target

Now the diffusion process itself is associated with the transformation.
The idea becomes:

$$\boxed{x \rightarrow \text{intermediate states} \rightarrow y} \quad (238)$$

rather than:

$$\boxed{\text{noise} \rightarrow y \quad \text{conditioned on } x} \quad (239)$$

63.4 The "Blank Canvas" Intuition

Imagine I tell two artists:

63.4.1 Artist A

Here is a completely blank canvas. Here is a sketch of the object. Now paint the final realistic image while looking at the sketch.

63.4.2 Artist B

Here is the sketch. Start modifying this sketch until it becomes the realistic image.

Artist B has a huge advantage:

63.4.3 Artist A:

blank

↓

???

↓

???

↓

realistic image

63.4.4 Artist B:

sketch

↓

preserve structure

↓

add details

↓

add texture

↓

realistic image

The source already contains information. Why throw that information away and start from completely random noise?

That's the basic intuition.

63.5 But There's an Important Correction

You cannot simply do:

$$\text{noise} = x_{\text{source}} \quad (240)$$

because the source is not Gaussian noise.

Instead, you can think of the source as defining the starting state or trajectory, depending on the particular diffusion formulation.

For example, one can construct a noisy version of the source:

$$x_t = \alpha_t x_0 + \sigma_t \epsilon \quad (241)$$

where:

$$x_0 = x_{\text{source}} \quad (242)$$

Then the model learns to move through states toward the target.

So conceptually:

source image

add controlled noise

x_t

learned transformation

target

The exact mathematical construction depends on the paper.

63.6 So Why is This Better if Both Can Produce Accurate Results?

This is the really important question.

It isn't necessarily better in every situation. A conditional diffusion model can absolutely produce:

$$\text{source} \rightarrow \text{accurate target} \quad (243)$$

because the network can learn $p(y|x)$.

The problem is more about what the model is being asked to learn.

63.6.1 Conditional approach

The model has to learn: From random noise, construct a plausible target while using x to constrain the result. That's a difficult generation problem.

63.6.2 Source-to-target trajectory

The model has something closer to: Start with something that already resembles the source and progressively transform it toward the target. This can make the relationship between source and target more explicit.

63.7 Here's a Concrete Example

63.7.1 Source:

rough human face

_---/

63.7.2 Target:

realistic human face

The source already tells you:

- where the eyes are
- where the nose is
- where the mouth is
- approximate shape
- spatial arrangement

63.7.3 Conditional diffusion

Start: random noise

Then:

noise
↓
vague structure
↓
face-like structure
↓
face
↓
realistic face

The model has to create the structure while being guided by the source.

63.7.4 Source-based trajectory

Start: rough face

Then:

rough face
↓
better face structure
↓
skin structure
↓
texture
↓
lighting
↓
realistic face

The model doesn't need to discover from scratch: "Where should the eyes go?" That information is already there.

63.8 And This Connects to the Domain Issue

Now we can bring back the "domain" terminology.

Suppose:

$$x \in X \tag{244}$$

where X = sketch images. And:

$$y \in Y \quad (245)$$

where $Y = \text{photographs}$.

These are two different distributions:

$$p_X(x) \neq p_Y(y) \quad (246)$$

The translation problem is:

$$X \rightarrow Y \quad (247)$$

A normal conditional diffusion model primarily learns the target distribution:

$$p_Y(y) \quad (248)$$

and uses x to condition generation:

$$p_Y(y|x) \quad (249)$$

The source does not necessarily define a diffusion path from distribution X to distribution Y .

The criticism is that for translation, it can be desirable to explicitly model:

$$\boxed{X \rightarrow Y} \quad (250)$$

rather than:

$$\boxed{\text{noise} \rightarrow Y \quad \text{with } X \text{ as a condition}} \quad (251)$$

63.9 One More Subtle Point: "Starting from Source" Doesn't Mean Preserve Everything

This is important.

If you simply add a little noise to the source and denoise it, you might get:

$$\text{source} \rightarrow \text{almost same source} \quad (252)$$

That's not necessarily the desired target.

For example: sketch $\rightarrow$ photograph requires the model to change the domain.

So the goal isn't: "Don't change the source." It is: "Use the source's existing information as the starting point, while allowing the diffusion process to transform its representation into the target domain."

So:

$$\boxed{\text{preserve useful information} + \text{change domain}} \quad (253)$$

That's the sweet spot.

63.10 The Intuition in One Sentence

Yes: If the task is truly translation, it can be more natural to transform an existing structured source image into the target than to generate the target from random noise and merely tell the generator what the source looks like.

But don't interpret the paper as saying: "Conditional diffusion is bad." It's saying something more specific: Conditional diffusion treats translation as conditional generation, whereas the authors want diffusion itself to model the source-to-target transformation.

And that's why their criticism is subtle: both approaches can generate good images, but they formulate the underlying problem differently.

64 The Complete Picture: Training vs Inference

In diffusion/BBDM training, there are two things happening:

64.1 Generate the Training State Using the Forward Bridge

We start with the known training pair:

$$(x_0, y) \tag{254}$$

Choose a random t , sample ϵ , and construct:

$$\boxed{x_t = (1 - m_t)x_0 + m_t y + \sqrt{\delta_t} \epsilon} \tag{255}$$

So yes, this is data/state generation for training. We are not learning this equation; it is already defined.

64.2 Learn the Reverse Process

Now give the generated x_t to the neural network:

$$x_t, t \rightarrow \epsilon_\theta(x_t, t) \tag{256}$$

and compare its prediction with the training target:

$$\boxed{v_t = m_t(y - x_0) + \sqrt{\delta_t} \epsilon} \tag{257}$$

The loss updates the neural network parameters. So the network learns the information required to perform the reverse transition:

$$\boxed{x_t \rightarrow x_{t-1}} \tag{258}$$

64.3 Therefore

You can think of training as:

$$\boxed{\text{Known } (x_0, y) \xrightarrow{\text{forward bridge}} x_t \xrightarrow[\text{loss}]{\text{network}} \text{learn reverse dynamics}} \tag{259}$$

Then after training:

$$\boxed{x_T \approx y \rightarrow x_{T-1} \rightarrow \cdots \rightarrow x_0} \quad (260)$$

64.4 One Important Wording

I would say: Training uses the forward process to generate noisy/intermediate training states and uses those states to train a neural network that parameterizes the reverse process.

Rather than simply saying "training means data generation." Because data generation is only one part of training; the important learning part is the optimization of the neural network for the reverse process.

65 Summary of Key Equations

This section collects all the key equations from the paper.

Table 2: Key Equations in Brownian Bridge Diffusion Models

Equation	Description
$p(x_t x_0, x_T) = \mathcal{N}((1 - \frac{t}{T})x_0 + \frac{t}{T}x_T, \frac{t(T-t)}{T}I)$	Equation (3): Brownian Bridge
$q_{BB}(x_t x_0, y) = \mathcal{N}((1 - m_t)x_0 + m_ty, \delta_t I)$	Equation (4): Forward marginal
$\delta_t = 2s(m_t - m_t^2)$	Equation (5): Variance schedule
$x_t = (1 - m_t)x_0 + m_ty + \sqrt{\delta_t}\epsilon$	Equation (6): Reparameterized bridge
$\epsilon \sim \mathcal{N}(0, I)$	Gaussian noise
$q_{BB}(x_t x_{t-1}, y) = \mathcal{N}(a_tx_{t-1} + b_ty, \delta_{t t-1}I)$	Equation (8): Transition distribution
$p_\theta(x_{t-1} x_t, y) = \mathcal{N}(\mu_\theta(x_t, t), \tilde{\delta}_t I)$	Equation (9): Reverse process
$\mu_\theta = c_{xt}x_t + c_{yt}y + c_{\epsilon t}\epsilon_\theta$	Equation (14): Reverse mean
$x_{t-1} = c_{xt}x_t + c_{yt}y - c_{\epsilon t}\epsilon_\theta(x_t, t) + \sqrt{\tilde{\delta}_t}z$	Sampling update
$x_T = y$	Endpoint initialization
$x_0 = x$	Source initialization

66 Conclusion

This comprehensive survey has covered the complete mathematical and architectural framework of Brownian Bridge Diffusion Models (BBDM), including:

1. **The Problem:** Image-to-image translation between two domains $A \rightarrow B$ where $x \in A$ and $y \in B$.

2. **The Traditional Approach:** Conditional generation where noise $\rightarrow$ target with source as condition.
3. **The Criticism:** Source is outside the diffusion trajectory; the model learns noise $\rightarrow$ target with source as auxiliary information.
4. **The Central Idea:** Construct a stochastic Brownian Bridge where $x_0 = x$ and $x_T = y$, so the diffusion trajectory itself represents the domain relation.
5. **The Bridge Equation:** $x_t = (1 - m_t)x_0 + m_ty + \sqrt{\delta_t}\epsilon$ - the heart of BBDM.
6. **The Variance Schedule:** $\delta_t = 2s(m_t - m_t^2)$ with parameter s controlling diversity.
7. **The Reverse Process:** $x_T = y \rightarrow x_{T-1} \rightarrow \dots \rightarrow x_0$ with network predicting the bridge displacement.
8. **The Key Distinction:** Conditional diffusion treats source as condition; BBDM treats source and target as endpoints of the stochastic bridge.
9. **The Intuition:** Starting from an already structured source image is more natural for translation than starting from random noise.

The deepest idea of the paper is:

Instead of asking diffusion to generate B conditioned on A, BBDM constructs a stochastic diffusion b

(261)

That is why the authors call it direct domain-to-domain translation rather than conditional generation.

BBDM = stochastic diffusion process connecting two endpoints

(262)